

Linear Regression with Uber and Lyft Ride Data

1 Introduction

In this assignment, I will explore and apply linear regression techniques to model and predict ride prices using a real-world dataset from Uber and Lyft services in Boston, MA. The dataset contains hundreds of thousands of ride entries with various contextual features. The goal is to build and evaluate regression models that estimate ride prices accurately and to investigate which types of linear regression approaches yield the best results.

2 Data

2.1 Dataset

This dataset contains a large number of ride records, each with a diverse set of attributes. It is composed of 57 columns and 693071 entries.

2.2 Features

There are 57 feature columns. Using these features we can uncover insights and predict ride prices.

The features are as follows:

	Column	Non-Null Count	Dtype
0	id	693071 non-null	object
1	timestamp	693071 non-null	float64
2	hour	693071 non-null	int64

3	day	693071 non-null int64
4	month	693071 non-null int64
5	datetime	693071 non-null object
6	timezone	693071 non-null object
7	source	693071 non-null object
8	destination	693071 non-null object
9	cab_type	693071 non-null object
10	product_id	693071 non-null object
11	name	693071 non-null object
12	price	637976 non-null float64
13	distance	693071 non-null float64
14	surge_multiplier	693071 non-null float64
15	latitude	693071 non-null float64
16	longitude	693071 non-null float64
17	temperature	693071 non-null float64
18	apparentTemperature	693071 non-null float64
19	short_summary	693071 non-null object
20	long_summary	693071 non-null object
21	precipIntensity	693071 non-null float64
22	precipProbability	693071 non-null float64
23	humidity	693071 non-null float64
24	windSpeed	693071 non-null float64
25	windGust	693071 non-null float64
26	windGustTime	693071 non-null int64
27	visibility	693071 non-null float64
28	temperatureHigh	693071 non-null float64
29	temperatureHighTime	693071 non-null int64
30	temperatureLow	693071 non-null float64
31	temperatureLowTime	693071 non-null int64

32	apparentTemperatureHigh	693071 non-null float64
33	apparentTemperatureHighTime	693071 non-null int64
34	apparentTemperatureLow	693071 non-null float64
35	apparentTemperatureLowTime	693071 non-null int64
36	icon	693071 non-null object
37	dewPoint	693071 non-null float64
38	pressure	693071 non-null float64
39	windBearing	693071 non-null int64
40	cloudCover	693071 non-null float64
41	uvIndex	693071 non-null int64
42	visibility.1	693071 non-null float64
43	ozone	693071 non-null float64
44	sunriseTime	693071 non-null int64
45	sunsetTime	693071 non-null int64
46	moonPhase	693071 non-null float64
47	precipIntensityMax	693071 non-null float64
48	uvIndexTime	693071 non-null int64
49	temperatureMin	693071 non-null float64
50	temperatureMinTime	693071 non-null int64
51	temperatureMax	693071 non-null float64
52	temperatureMaxTime	693071 non-null int64
53	apparentTemperatureMin	693071 non-null float64
54	apparentTemperatureMinTime	693071 non-null int64
55	apparentTemperatureMax	693071 non-null float64
56	apparentTemperatureMaxTime	693071 non-null int64

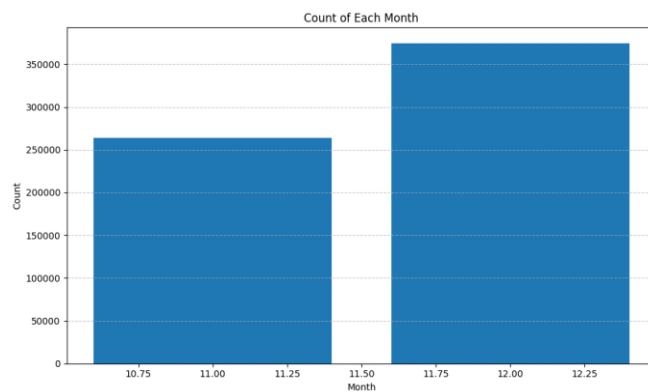
Here we have 55095 missing values only in “price” col. The best way to handle missing values in here is to drop them. Since all of these missing values only appear in “ price” column and we want to predict this feature, so with no value for them, they will not be useful here.

Additionally, we also have duplicate cols (“visibility” and “visibility.1”) which we must drop one and only keep one of them.

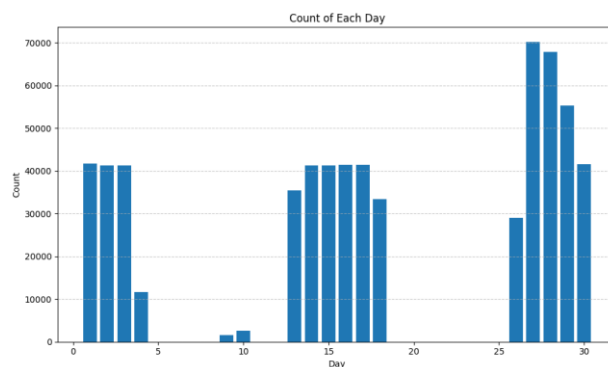
2.3 Analysis

In this section we will go through some analyses and visualizations related to the data that can help us to have a better sense of our data and the relationship between the features.

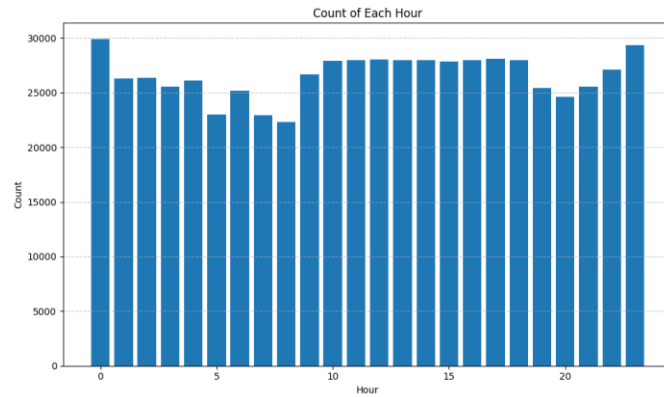
Time Analysis:



It appears that we only have november and december in our month data. It means the data is only recorded or taken in november and december with december data dominating.

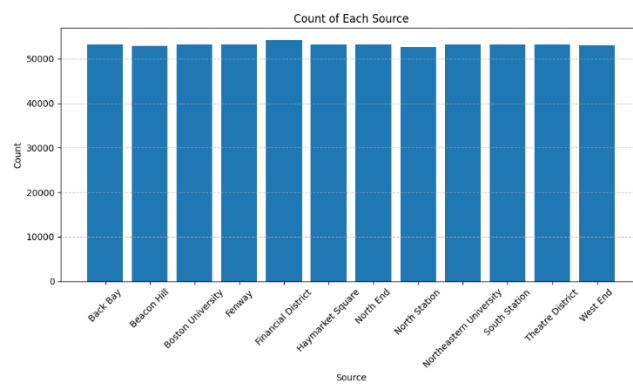


It seems we have many gaps in our 'day' data. For example we don't have data from 18th day until 25th day in each month.

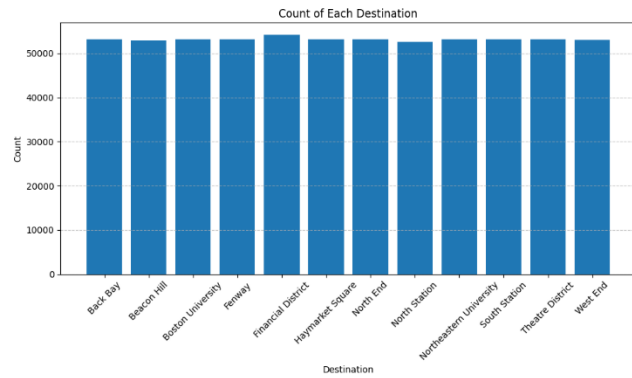


It seems we have almost 24 hours recorded data.

Source and Destination Analysis:

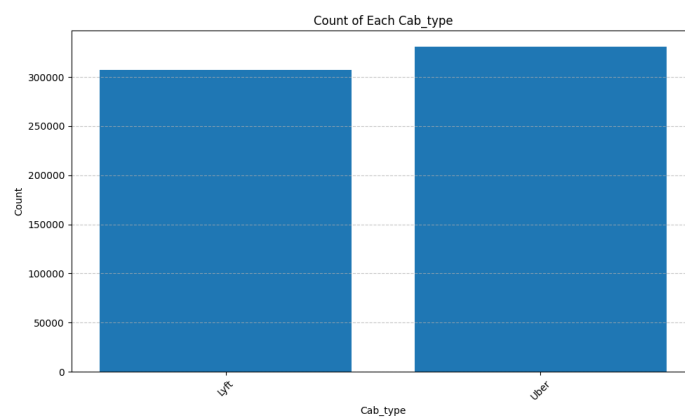


It seems that all sources are almost equal in number. There are about 50k data in each source feature (Back Bay, Beacon Hill, Boston University, etc).

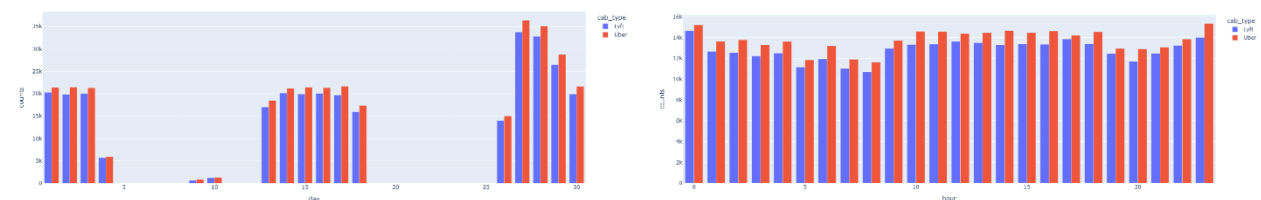


Same with source feature, there are about 50k data in each destination feature (Back Bay, Beacon Hill, Boston University, etc).

Cab Type Analysis:

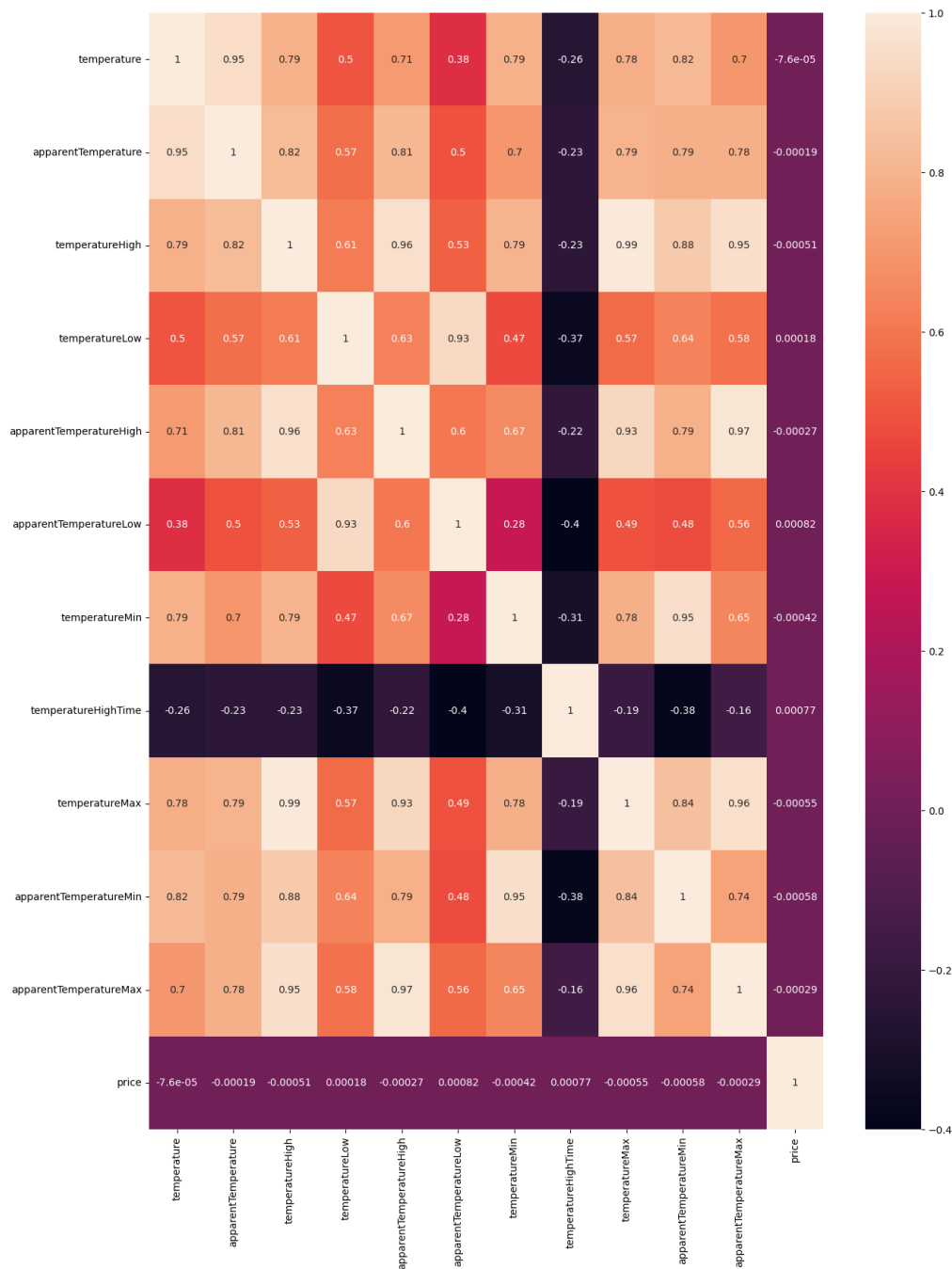


So for our whole data, we have uber data more than lyft data. The difference is not too big, each cab type has about 300K data.

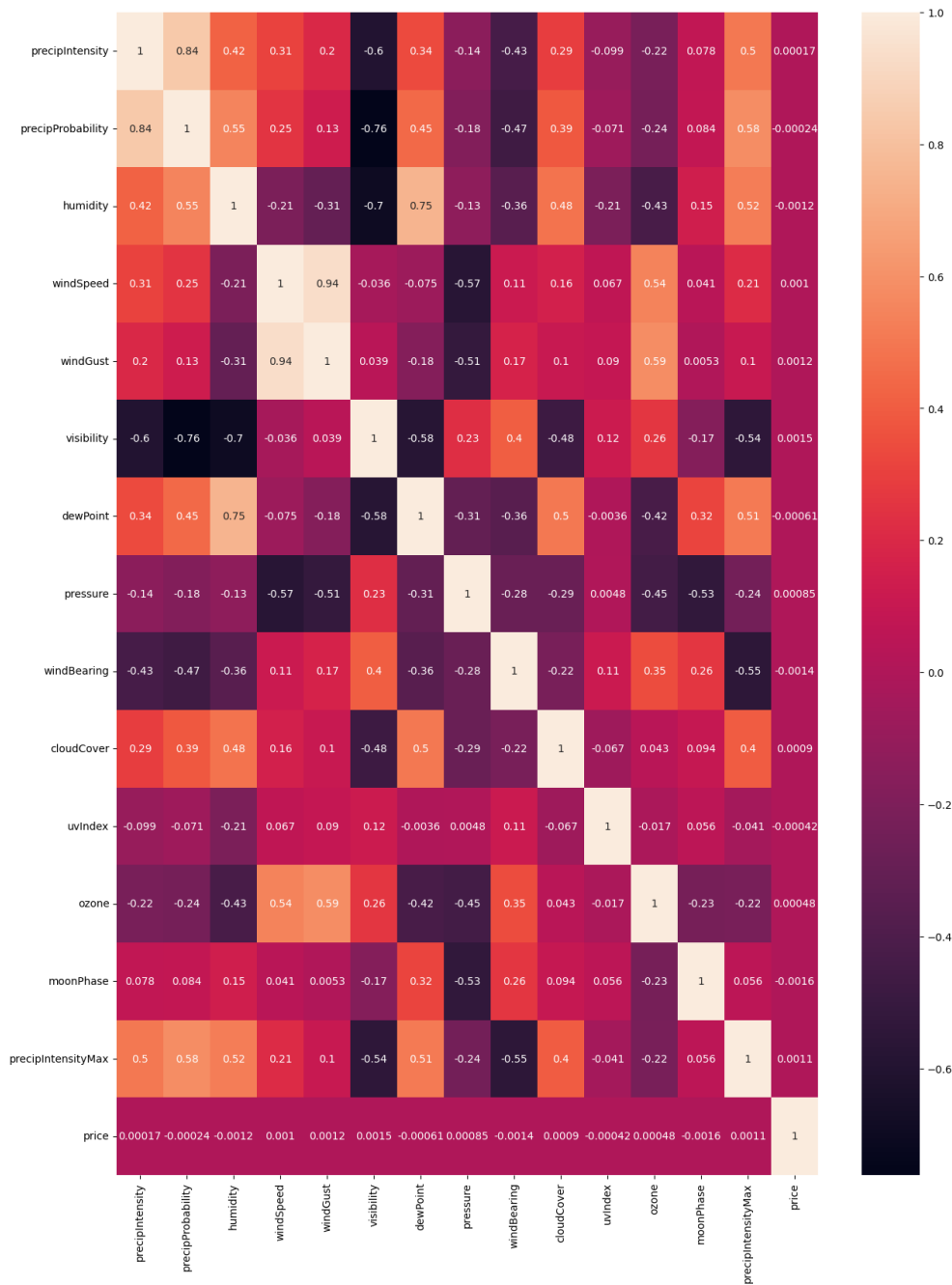


So in every day and every hour recorded, uber seems dominating booking order in our data.

Price Analysis:



We see that all temperature related features have weak correlation with our target feature which is price. Removing all of them will not make any impact to our regression model.



Apparently all climate related features also have weak correlation with our target feature which is price. Once again, removing all of them will not make any impact to our regression model.

i want to check our categorical value in our dataset features:

timezone : ['America/New_York']

source : ['Haymarket Square' 'Back Bay' 'North End' 'North Station' 'Beacon Hill'
'Boston University' 'Fenway' 'South Station' 'Theatre District'
'West End' 'Financial District' 'Northeastern University']

destination : ['North Station' 'Northeastern University' 'West End' 'Haymarket Square'
'South Station' 'Fenway' 'Theatre District' 'Beacon Hill' 'Back Bay'
'North End' 'Financial District' 'Boston University']

cab_type : ['Lyft' 'Uber']

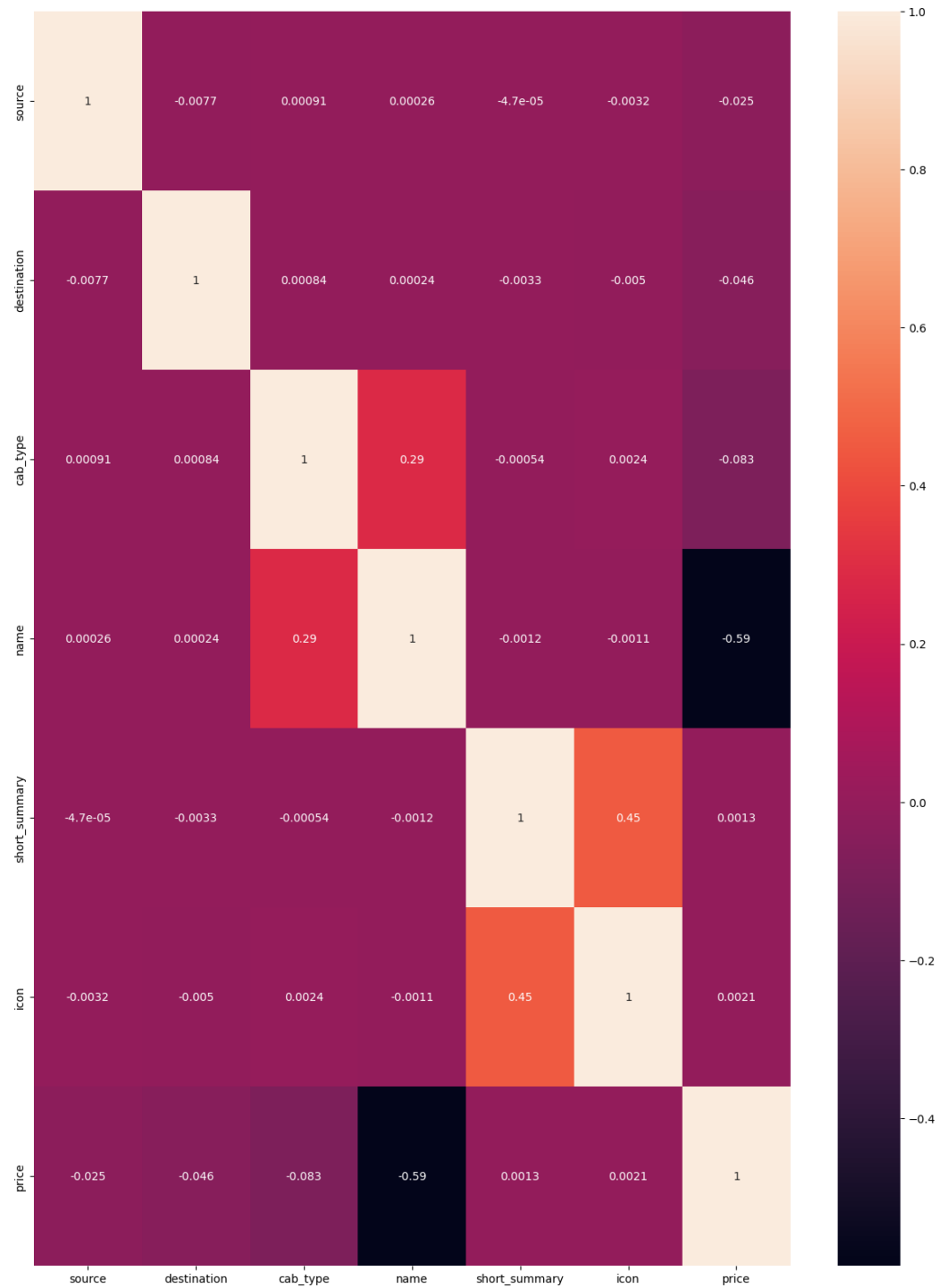
product_id : ['lyft_line' 'lyft_premier' 'lyft' 'lyft_luxsuv' 'lyft_plus' 'lyft_lux'
'6f72dfc5-27f1-42e8-84db-ccc7a75f6969'
'6c84fd89-3f11-4782-9b50-97c468b19529'
'55c66225-fbe7-4fd5-9072-eab1ece5e23e'
'9a0e7b09-b92b-4c41-9779-2ad22b4d779d'
'6d318bcc-22a3-4af6-bddd-b409bfce1546'
'997acbb5-e102-41e1-b155-9df7de0a73f2']

name : ['Shared' 'Lux' 'Lyft' 'Lux Black XL' 'Lyft XL' 'Lux Black' 'UberXL'
'Black' 'UberX' 'WAV' 'Black SUV' 'UberPool']

short_summary : [' Mostly Cloudy ' ' Rain ' ' Clear ' ' Partly Cloudy ' ' Overcast '
' Light Rain ' ' Foggy ' ' Possible Drizzle ' ' Drizzle ']

icon : [' partly-cloudy-night ' ' rain ' ' clear-night ' ' cloudy ' ' fog '
' clear-day ' ' partly-cloudy-day ']

We can see that 'timezone' feature has only 1 value and 'product_id' feature contains many unidentified values. So we can remove or drop them.



We can see only name feature that has a relatively strong correlation. Source, destination, and cab_type features have relatively weak correlation, but i will pick cab_type feature because it has stronger correlation than other two features. I will drop or remove the rest of the columns.

Also i will remove hour, day, month, latitude, longitude, because we won't need them for now.

The final remaining cols that have an impact on price:

```
Index(['cab_type', 'name', 'price', 'distance', 'surge_multiplier'], dtype='object')
```

Then I check the outliers and remove them.

we have 5588 data outliers. We can remove or drop them.

Size before removing : (637975, 5)

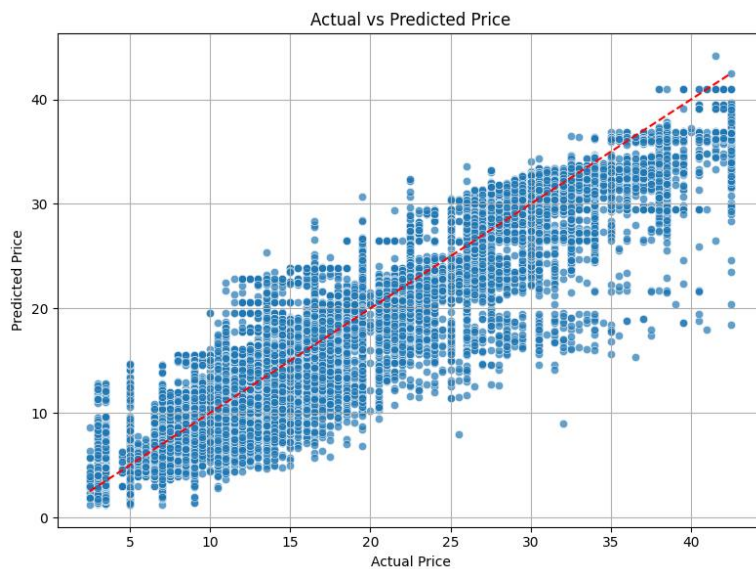
Size after removing : (632387, 5)

3 Modeling

In this step, first of all I performed the Encoding Data with One Hot Encoding approach.

I have also already handled the outliers.

3.1 Baseline Linear Regression



Mean Absolute Error (MAE): 1.67

Mean Squared Error (MSE): 5.08

R^2 (Coefficient of Determination): 0.9343

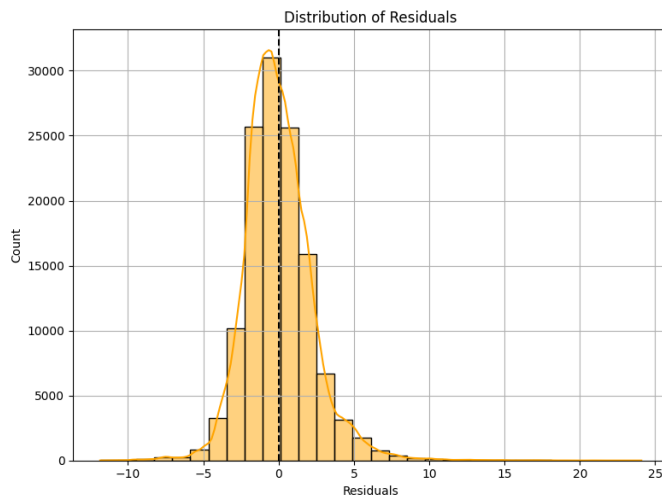
we understand that:

The model is accurate and generalizes well on the test set.

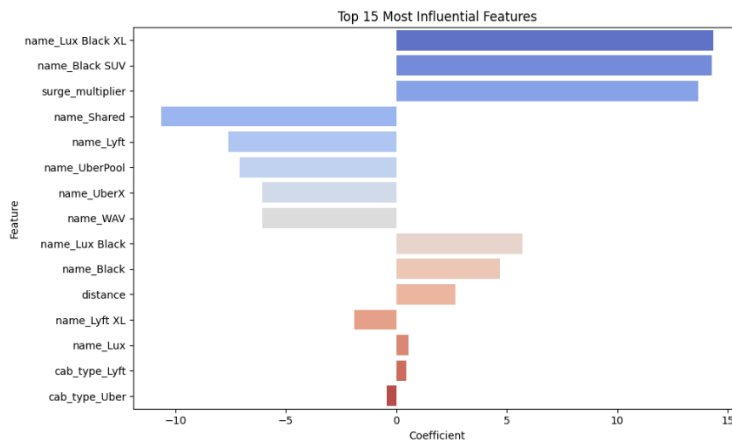
Errors are small on average, and prediction variance aligns closely with actual data.

If this is test set performance, it likely indicates no major overfitting and that features are well-explained by a linear relationship.

we still can improve it with other regression models which could give better results.



The plot of the distribution of residuals.



The plot of the top 15 influential features.

3.2 Ridge

First we try to find the best alpha using cross validation.

Best alpha for Ridge: 1.62975

Ridge Regression Evaluation:

MAE: 1.67

MSE: 5.08

R^2 : 0.9343

As we can see the evaluation metrics are the same to the baseline model.

3.3 Lasso

Again we use cross validation and the best alpha for Lasso will be: 0.00100

Lasso Regression Evaluation:

MAE: 1.67

MSE: 5.08

R^2 : 0.9342

As we can see the evaluation metrics are the same to the baseline model

We conclude that the results from Ridge, Lasso and the baseline model are the same.

3.4 Polynomial (degree = 2)

Polynomial Regression (degree 2) Results:

MAE: 1.26

MSE: 3.19

R^2 : 0.9587

we see a little bit of improvement but not very considerable.

3.5 Trade-Offs

1. Model Complexity

Linear Regression is most basic it can only fit linear lines.

Ridge/Lasso add regularization and lower complexity by penalizing coefficients (particularly Lasso, which can set coefficients to zero).

Polynomial Regression has much greater complexity it more features and chances of overfitting.

2. Performance

based on the metrics, (MAE, MSE, R^2) :

Polynomial regression has minimum error but is likely to overfit.

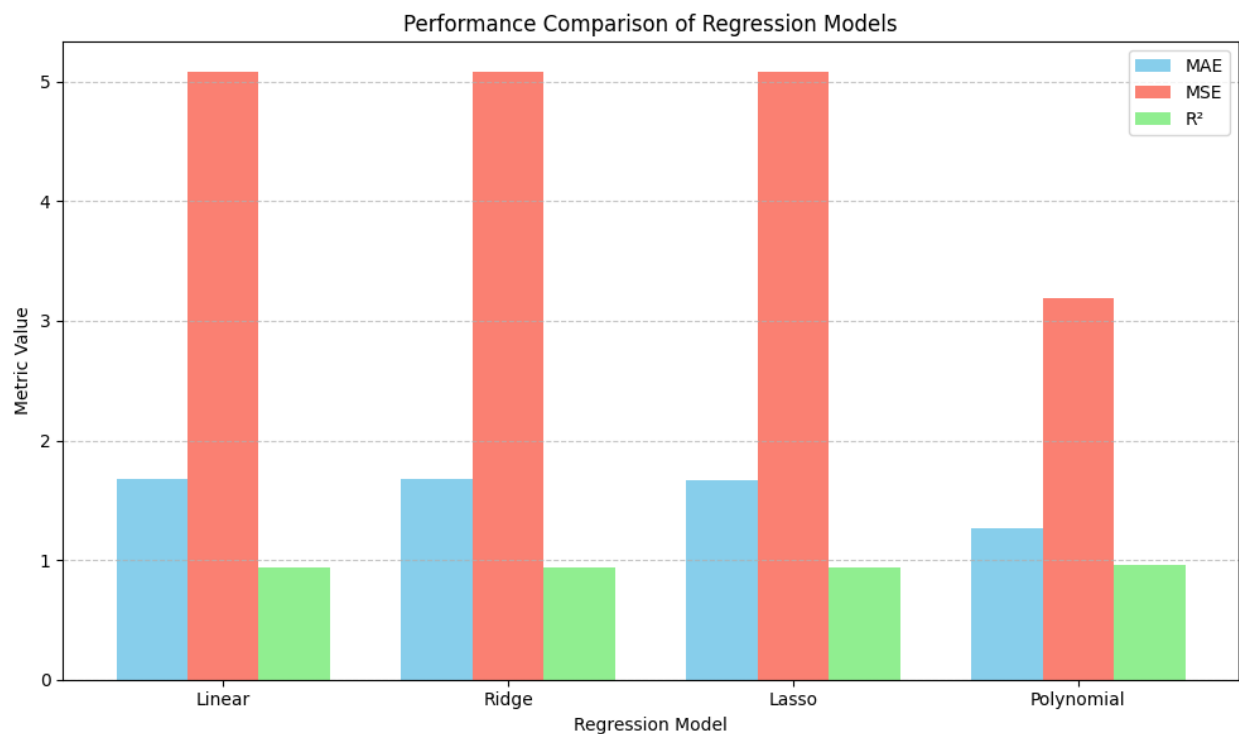
Ridge and Lasso offer small performance gain in general but in here they were almost identical with the baseline model and they also handle complexity.

3. Interpretability

Linear Regression: Very interpretable its coefficients represent simple feature effect.

Ridge/Lasso: Coefficients are shrunk or dropped, so Lasso is especially well-adapted to feature selection.

Polynomial Regression: Low interpretability its interactions and squared terms make it difficult to explain to stakeholders.



Metric		Observation
MAE (Mean Absolute Error)	Polynomial Regression has the lowest MAE, indicating better average prediction accuracy than the others. Linear, Ridge, and Lasso have nearly identical and higher MAE values.	
MSE (Mean Squared Error)	Polynomial Regression significantly reduces the MSE, suggesting it handles outliers or large errors better than the linear models. The other three models have similar and notably higher MSEs.	
R ² (Coefficient of Determination)	All models have very similar and high R ² scores (close to 1), suggesting they all explain a similar proportion of the variance in the data, but Polynomial is slightly better.	
Aspect	Polynomial Regression	Linear/Ridge/Lasso
Complexity	Higher (due to added polynomial features)	Lower (simpler models)
Performance	Better (lower MAE/MSE, higher R ²)	Worse (especially in MSE)
Interpretability	Lower (harder to interpret higher-degree terms)	Higher (coefficients directly tied to features)
Overfitting Risk	Increased if degree is too high	Lower, especially with Ridge/Lasso regularization

4 Model Enhancement through Data Preparation

In this step I performed the followings:

Target Encoding + Interaction Terms + Outlier Filtering (Isolation Forest)

Target Encoding (for cab_type and name):

What it do: Target encoding replace categorical variables with the mean of the target variable (price) for each category.

Performance: when i encode the categories as the mean price the model gets more informative features . It improve accuracy of prediction especially when certain categorical levels correlate well with target.

Interpretability: it makes the model simpler but it can reduce interpretability becausee it becomes harder to understand the meaning behind the encoded values compared to original categorical features.

Interaction Terms (between distance, surge_multiplier, cab_type_te, and name_te):

What it does: Interaction terms reveal the combined effect of two or more features on the target variable.

Performance: Including interaction terms can improve model performance by allowing it to focus on non-linear effects. If distance and surge_multiplier together have a more significant contribution to the price than if they were examined individually the interaction terms will allow the model to do a better job.

Generalization: Interaction terms increase the model's complexity which can help in generalization if the interactions are real in the data but can cause overfitting if they are spurious

Interpretability: More features (specifically interaction terms) can make the model harder to interpret. It is more difficult to understand the direct effect of each feature on the target .

Outlier Filtering (with Isolation Forest):

What it does: Isolation Forest is algorithm that detects and removes outliers by isolating anomalies in the data

Performance: Outliers are typically eliminated to improve model performance through a reduction in data noise.

Generalization: By removing noise from training data the model will generalize better data because it is aiming at central trends rather than being fitted to outliers.

Interpretability: This process make the model more focused but make the process less interpretable as we are removing parts of the dataset that have individual less frequent meaning.

Polynomial Regression (degree 2) Results (the model with the best performance among the models that we had so far):

MAE: 1.26

MSE: 3.42

R²: 0.9569

These result seem to be pretty good and the enhancements have done their job well but these enhancements seem to have weakened my original model a little bit (not considerable). I assume that the data preprocessings and the encoding that I used earlier before this section was so good that my data did not need any sort of new enhancement (because the evaluation metrics also were reasonable and good from the beginning).

Conclusion

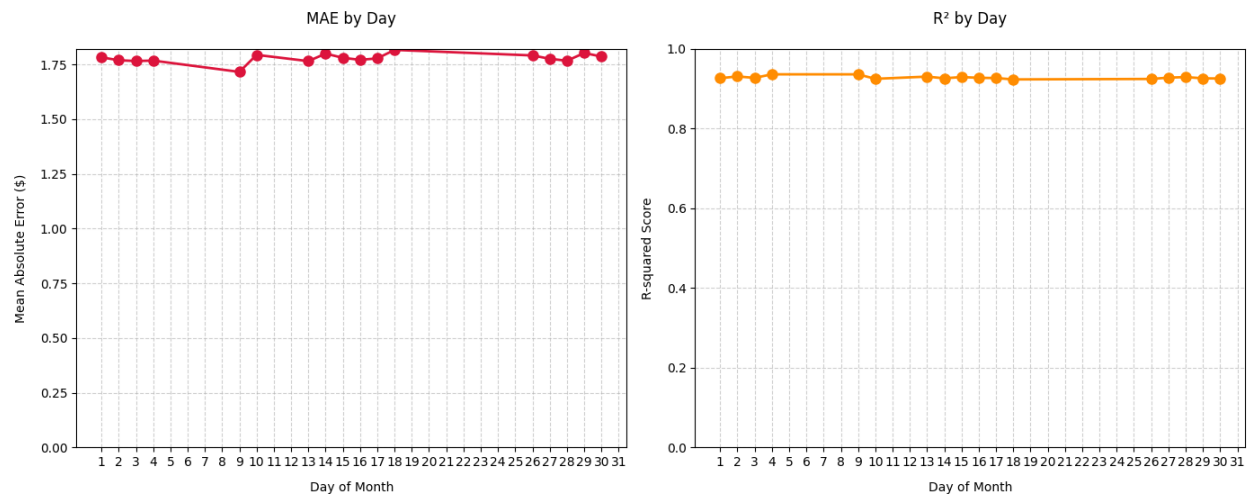
In this project, we conducted a deep analysis of Uber and Lyft Ride data to predict the price feature. Based on the analysis of different regression models, we can conclude that, Polynomial Regression outperforms the others in terms of error reduction and maintains high R². but it has less interpretability and can risk overfitting if not properly regularized or validated. For high stakes or interpretable scenarios Ridge or Lasso are better tradeoffs. For tasks that accuracy is more important and complexity is ok, Polynomial Regression is a good option.

Bonus

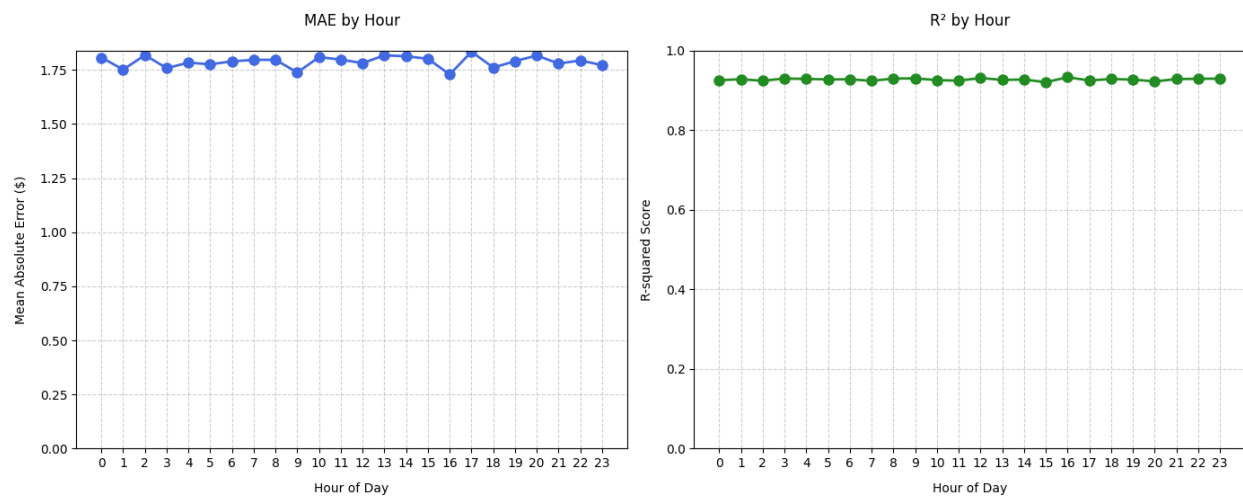
In this part I extracted hour and day information from timestamps. The code converts categorical data like ride types into numerical values using target encoding (replacing categories with their average prices). For the analyze it splits the data into hour and day segments training separate linear regression

models for each time segment with more than 100 data points. Each models performance is evaluated using two metrics: MAE to measure prediction accuracy and R^2 to assess how well the model explains price variations. The results are visualized in two pair plots one showing hourly performance (with hours 0-23) and another showing day performance (days 1-31) each displaying both MAE and R^2 trends

Model Performance by Day of Month



Model Performance by Hour of Day



References

[1] <https://www.kaggle.com/datasets/brillrb/uber-and-lyft-dataset-boston-ma>

Maryam Soleimani 401222075